VITISH 2026
(SIH Internal Hackathon)
TITLE PAGE

Problem Statement ID: 45
Problem Statement Title: AI-powered Radiology Interpretation Assistant
Theme: HealthTech
PS Category: Software

gv ant — Explainable Multi-Modality ec Support

❖ Proposed Solution (Describe your Idea / Solution / Prototype)

Detailed explanation of the proposed solution

How it addresses the problem

Innovation and uniqueness of the solution

Web CDS platform for X-ray, CT, MRI, and Ultrasound — radiologist assistant, not a replacement
Upload PNG/JPG/DICOM → modality router; DenseNet121 (CheXNet) predicts 18 chest X-ray findings
CT nodules, MRI tumor volume, US BI-RADS as live pipelines (nnU-Net / MONAI / MedSAM next)
Grad-CAM heatmaps + confidence + severity + ranked differential diagnosis (explainable AI)
Auto radiology report + PDF; FHIR ImagingStudy & DiagnosticReport; Orthanc PACS / DICOMweb
Low-confidence cases flagged 'Needs Human Review' — final interpretation by radiologist
Addresses India's radiologist shortage (~1 per 100,000 vs WHO norm 1:10,000)
Cuts rural reporting delays from 3–7 days to seconds for preliminary triage
Unique: prediction + evidence + report + PACS/FHIR/async worker — full hospital workflow

DenseNet121 (TorchXRayVision), Grad-CAM (Captum), Op

Backend: FastAPI, SQLAlchemy, SQLite, JWT + RBAC, Celery/Redis async DICOM worker
Frontend: React, Tailwind CSS, Chart.js dashboard, Cornerstone.js DICOM viewer
Imaging: pydicom, Orthanc PACS, DICOMweb QIDO-RS/WADO-RS, HL7 FHIR
Deploy: Docker Compose (API + Nginx + Redis + worker); optional Orthanc --profile pacs
Pipeline: Upload → encrypt → preprocess → modality AI → Grad-CAM → report → dashboard
Working prototype: XR CheXNet + CT/MRI/US demos, audit logs, encrypted images at rest

- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts / Images / working prototype)

Feasible: open-source stack, pretrained DenseNet121, Docker on modest hardware
MVP: full-stack XR CDS + live CT/MRI/US routes, PDF reports, PACS/FHIR hooks
Challenges: clinical validation, dataset bias, 3D CT/MRI weights, regulatory path
Risk: false negatives — confidence thresholds, Grad-CAM, radiologist-in-the-loop
Risk: modality mismatch / machine drift — DICOM tags + declared modality + CLAHE
Strategy: hospital pilot on X-ray first; attach nnU-Net / MONAI / MedSAM next

- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

Rural doctors: instant AI-assisted findings for X-ray/CT/MRI/US without waiting days
Radiologists: draft report ~30 min → ~30 sec; heatmap to verify AI reasoning
Patients: faster pneumonia/TB/nodule triage — 2,500 children die daily from pneumonia
Hospitals: less imaging backlog, PACS/FHIR-ready, lower tele-radiology cost
Social: equitable CDS in underserved areas; RBAC, audit logs, PHI encryption
Economic: India medical imaging ~$2.1B by 2028; tele-radiology growing ~23%/year

- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

Team built full-stack radiology AI (XR DenseNet + CT/MRI/US + PACS) in hackathon sprint
Domain: healthcare AI, multi-modality imaging, hospital interoperability (PACS/FHIR)
Deliverables: web app, Celery worker, Grad-CAM, PDF reports, FHIR APIs, Docker
Skills: Python/ML, React, FastAPI, DICOM/PACS/FHIR, Docker Compose
Proof: running prototype (docker compose up) — demo doctor@radiology.ai / demo1234

- Team's experience in the domain
- Why choosing the domain
- Proof of past work in the domain

TorchXRayVision DenseNet121: https://github.com/mlmed/torchxrayvision
Grad-CAM — Selvaraju et al.: https://arxiv.org/abs/1610.02391 | Captum: https://captum.ai/
NIH ChestX-ray14: https://nihcc.app.box.com/v/ChestXray-NIHCC | WHO workforce: https://www.who.int/
CT LIDC-IDRI + nnU-Net; MRI BraTS + MONAI; US BUSI + MedSAM (planned 3D models)
Orthanc PACS / DICOMweb: https://www.orthanc-server.com/
HL7 FHIR ImagingStudy + DiagnosticReport: https://www.hl7.org/fhir/

- Details / Links of the reference and research work